

Pouya Hatami

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EDUCATION

The University of British Columbia

Bachelor of Applied Science, Electrical Engineering, Dean's Honour Roll

Vancouver, BC

Sep. 2023 – May 2028

PROFESSIONAL EXPERIENCE

Kardium Inc.

FPGA Developer Intern

Burnaby, BC

May 2026 – Present

- Closed code-coverage gaps in AXI slave logic by creating a reusable, RAL-style VHDL test procedure library: each testbench classifies its registers by access type, and a single call exercises the shared AXI interface; integrated into every applicable testbench
- Designed an FPGA/firmware interlock for an I2C interface in VHDL through the Xilinx AXI IIC tri-state path and verified lockout behavior using assertions and Vivado ILA
- Implemented a firmware-controlled ADC reset architecture across five FPGA boards using AXI control, CDC synchronization, and MMCM/ODDR reset gating to block digital signals before the ADC power rails become stable
- Debugged and resolved AXI4 read-data corruption under RREADY backpressure across six slave peripherals; added RREADY-stall procedures to the shared AXI simulation package, raising coverage across five testbenches
- Added a Python static-analysis check enforcing a safety-critical synthesis acceptance criterion across two vendor-tool syntax variants
- Updated Python build automation used across the FPGA CI/CD pipeline to separate FSBL/PMUFW compilation from Vivado bitstream generation, reducing build time from approximately 60 to 10 minutes

RESEARCH EXPERIENCE

UBC ORCA Lab | Prof. Guy Lemieux

Undergraduate Research Assistant

Vancouver, BC

Dec. 2025 – May 2026

- Implemented execution-based run-control debugging on a single-cycle RV32I processor in SystemVerilog using the CV32E20 debug architecture as a reference, adding a debug FSM, four debug CSRs, `ebreak/dret` handling, and halt/resume PC redirection
- Documented the complete GDB–OpenOCD–RISC-V Debug Module workflow in an [open-source tutorial](#) covering processor halt, debug entry, program-buffer execution, and resume
- Verified halt–park–resume behavior, including `dpc` capture and restoration, using a self-checking Verilator testbench that emulated RISC-V Debug Module requests

PROJECTS

RV32I Pipelined Processor Core | *SystemVerilog, UVM*

Jun. 2026 – Present

- Designed and implemented a 5-stage pipelined RV32I processor with forwarding, hazard handling, M-mode CSRs, exceptions, timer interrupts, memory-mapped UART, and external halt/resume debug
- Verified the processor against a reference instruction-set simulator using UVM, with drivers, monitors, assertions, and a scoreboard comparing retired architectural state
- Verified privileged and debug control flow across the pipeline, including trap entry/return and halt/resume sequencing, with regression tests targeting pipeline hazards and flush behavior

LEADERSHIP EXPERIENCE

UBC Sailbot

Firmware Lead, Electrical Team

Vancouver, BC

Sep. 2024 – Present

- Led firmware development across UBC Sailbot's electrical team building POLARIS, a 3 m autonomous sailboat designed for multi-day Pacific data-collection missions
- Developed STM32 firmware for custom power-distribution hardware, implementing battery voltage and temperature monitoring, CAN-FD communication, and controlled shutdown/restart sequencing

SKILLS

RTL & Architecture: Verilog/SystemVerilog, VHDL, RISC-V, AXI4, Avalon-MM, CDC, timing analysis

Verification & Debug: UVM, SystemVerilog Assertions, ModelSim, Active-HDL, Vivado ILA, ISS-based verification

FPGA & EDA: Xilinx Zynq UltraScale+, Vivado, Vitis, Intel/Altera DE1-SoC, Quartus, ALINT-PRO

Programming & Embedded: C, C++, Embedded C, Python, Tcl, Assembly (8051, PicoBlaze), STM32, CAN-FD, Git, TeamCity CI